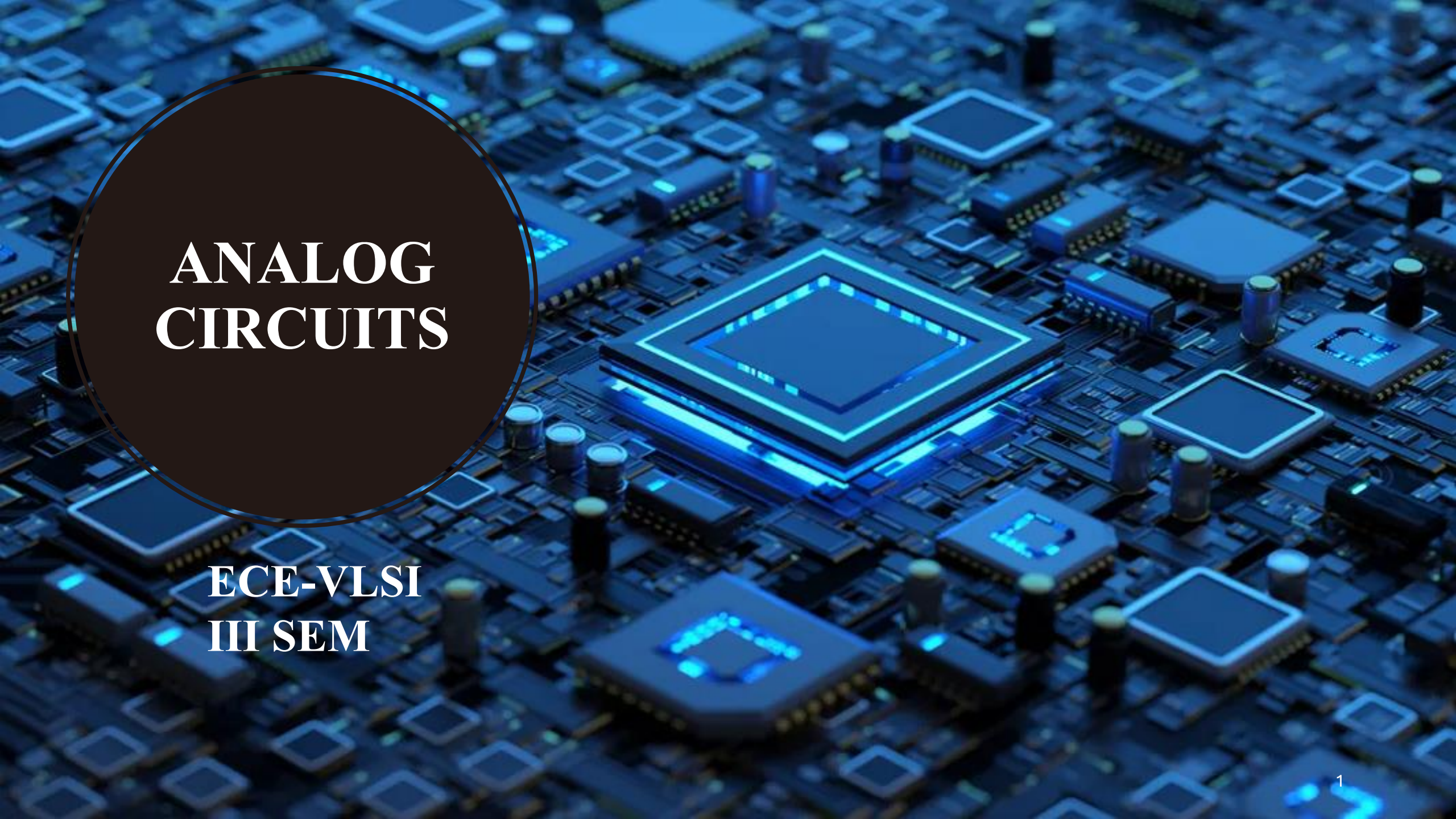
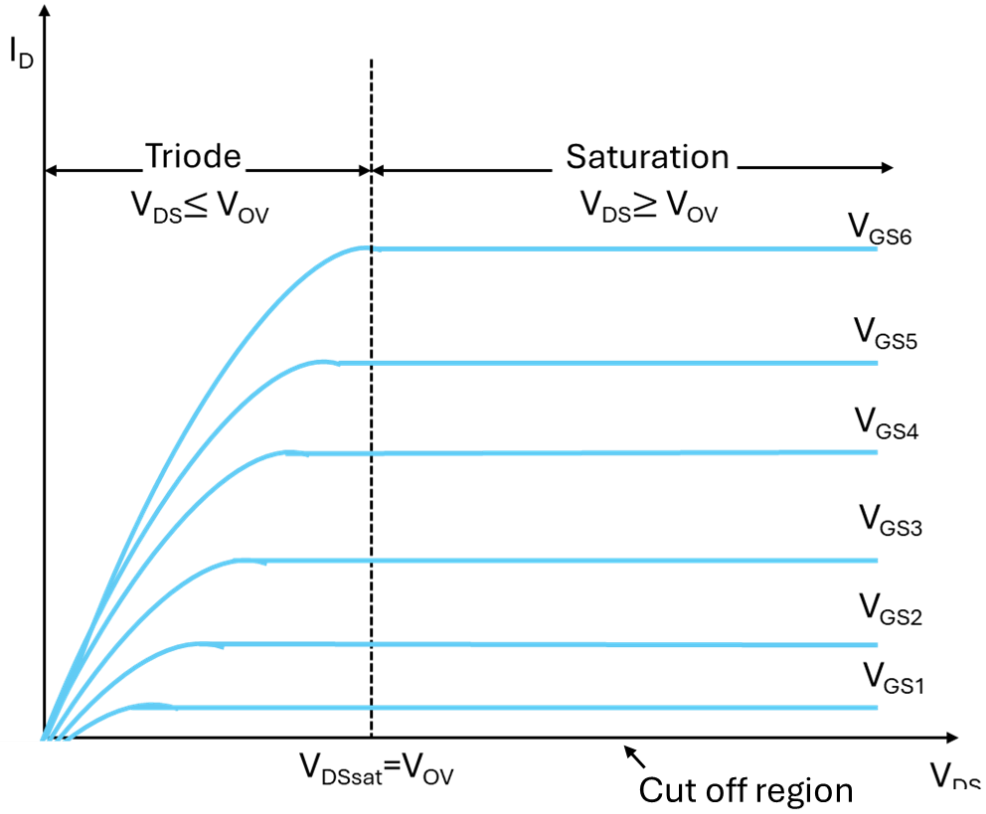
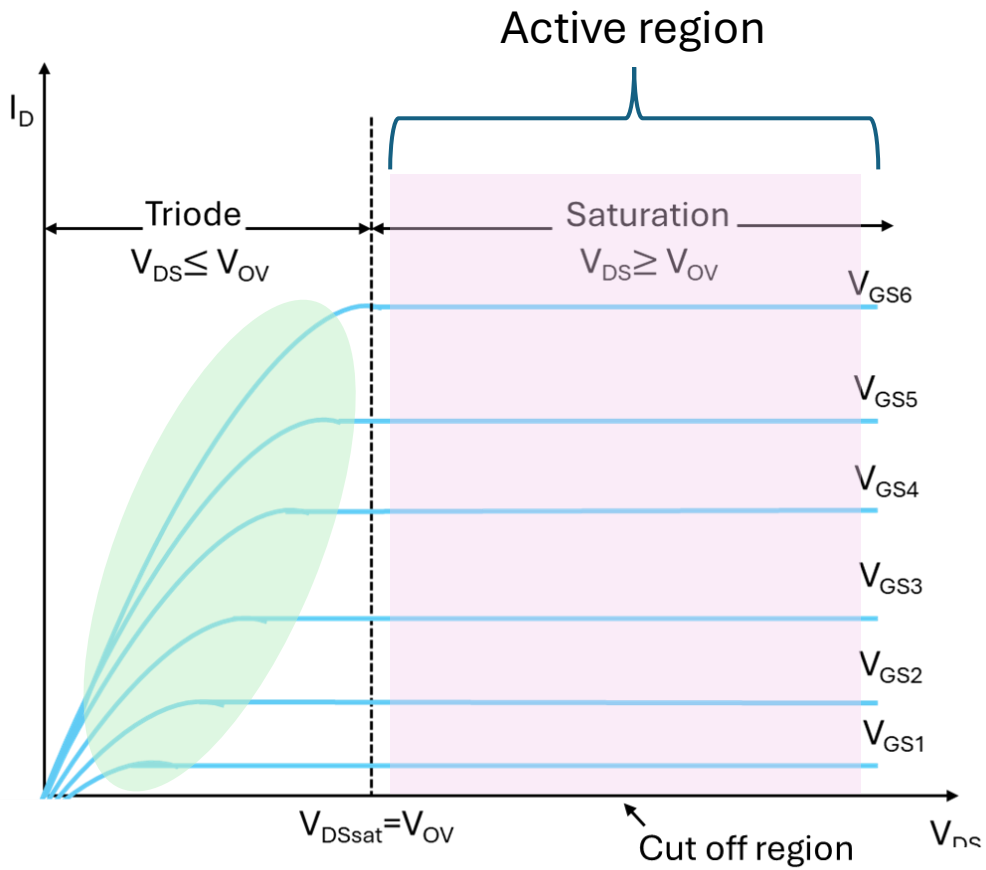

ANALOG
CIRCUITS
ECE-VLSI
III SEM

MOSFET BIASING

- Biasing required to set up the MOSFET operates in the **desired region (as switch, amplifier etc)**



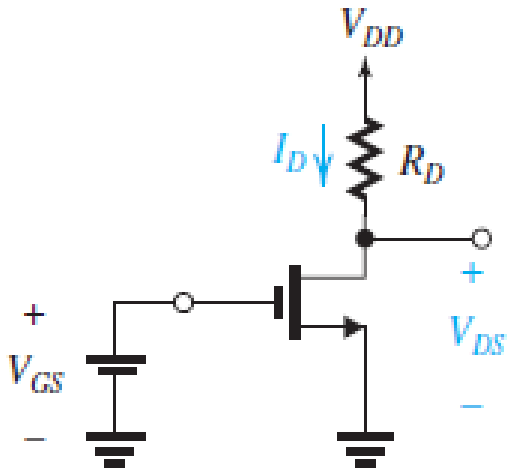


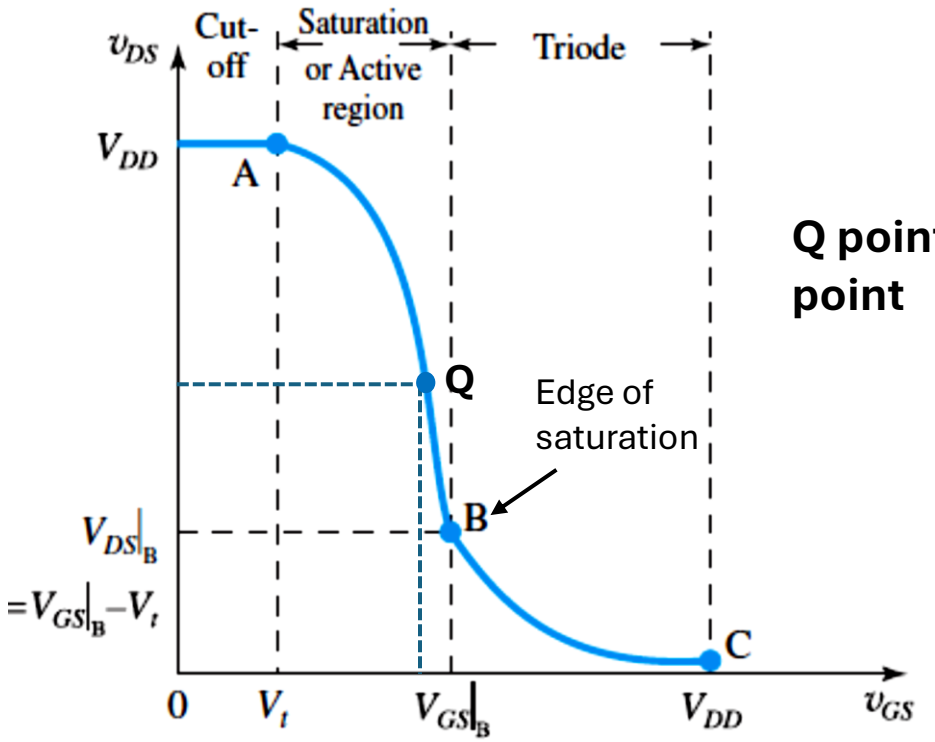
$$I_D = \frac{1}{2}\left[(\mu_n C_{ox})\left(\frac{W}{L}\right)\left((V_{GS} - V_t)^2\right)\right] \quad \text{(saturation)}$$

$I_D = f(V_{GS})$

MOSFET BIASING

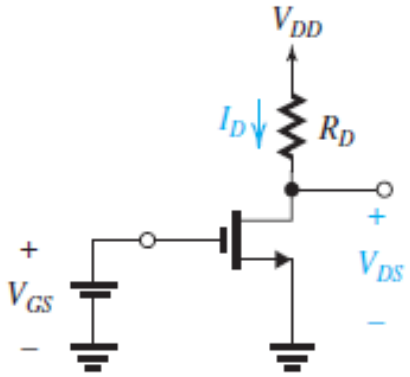
- In active region, MOSFET acts as a voltage-controlled current source
- Voltage between gate and source, V_{GS}, controls the drain current I_D
- Biasing enables linear amplification of MOSFET – amplification without distortion
- VTC- Voltage transfer characteristics – plot between output voltage versus the input voltage.

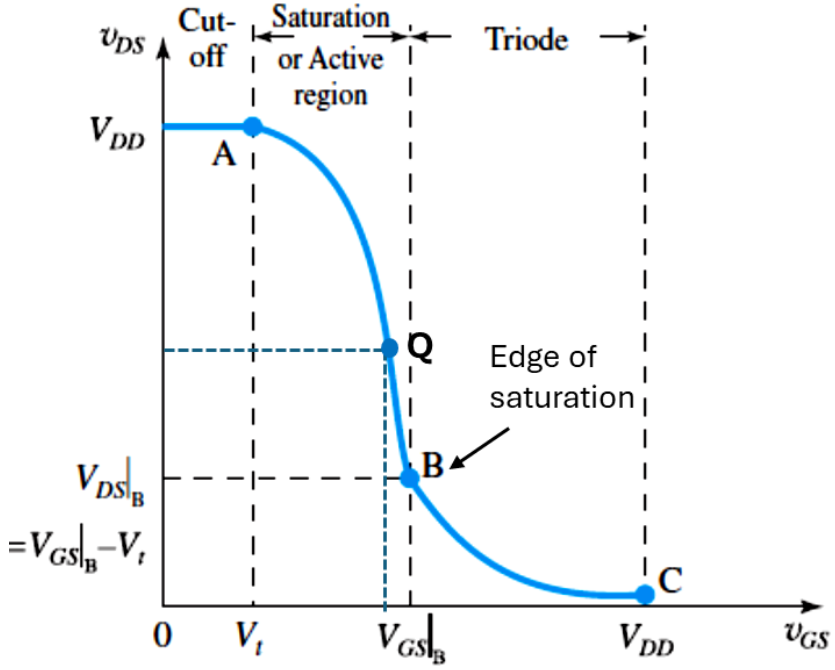


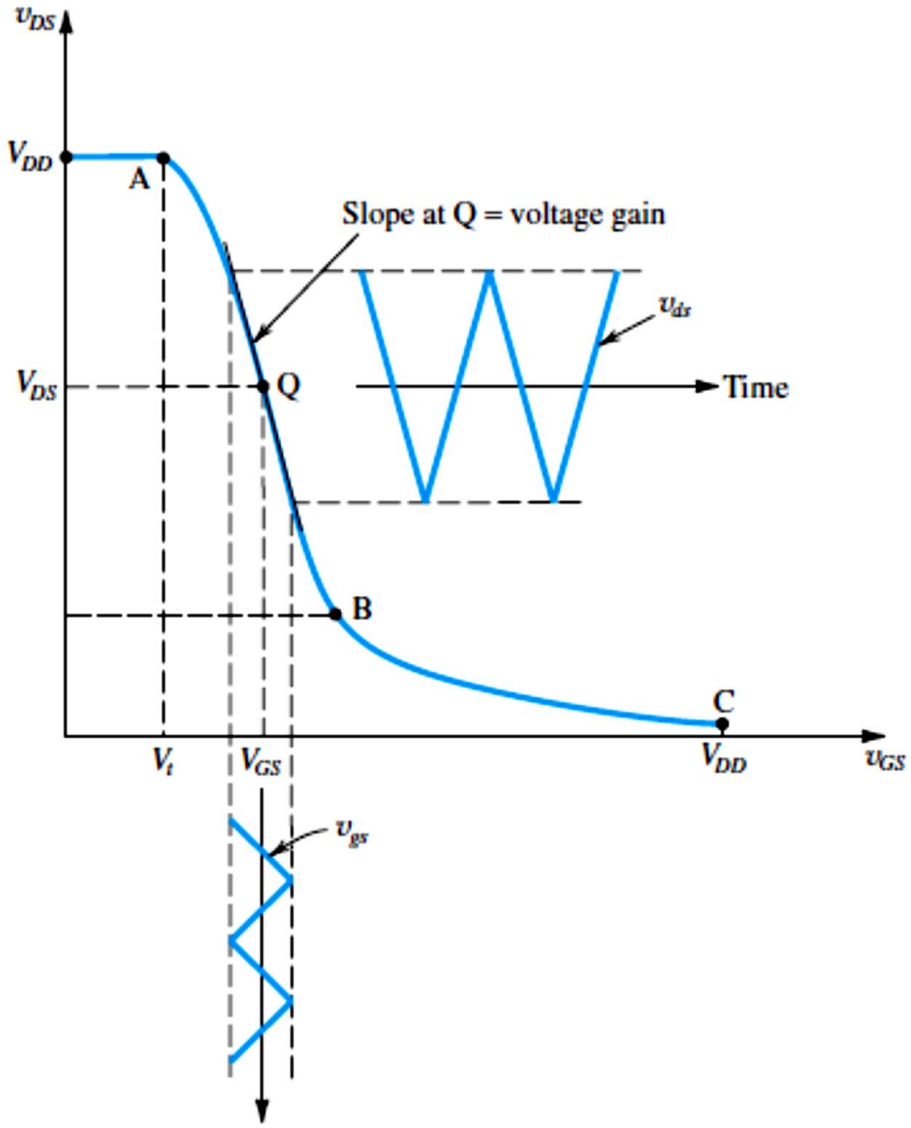


Q point- Quiescent point or DC operating point

MOSFET BIASING



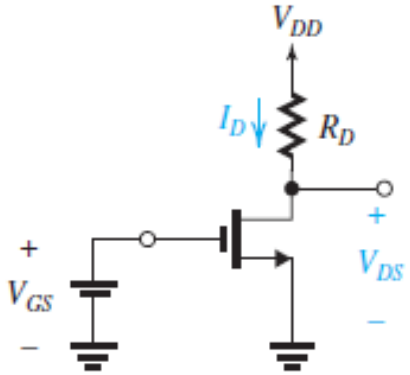


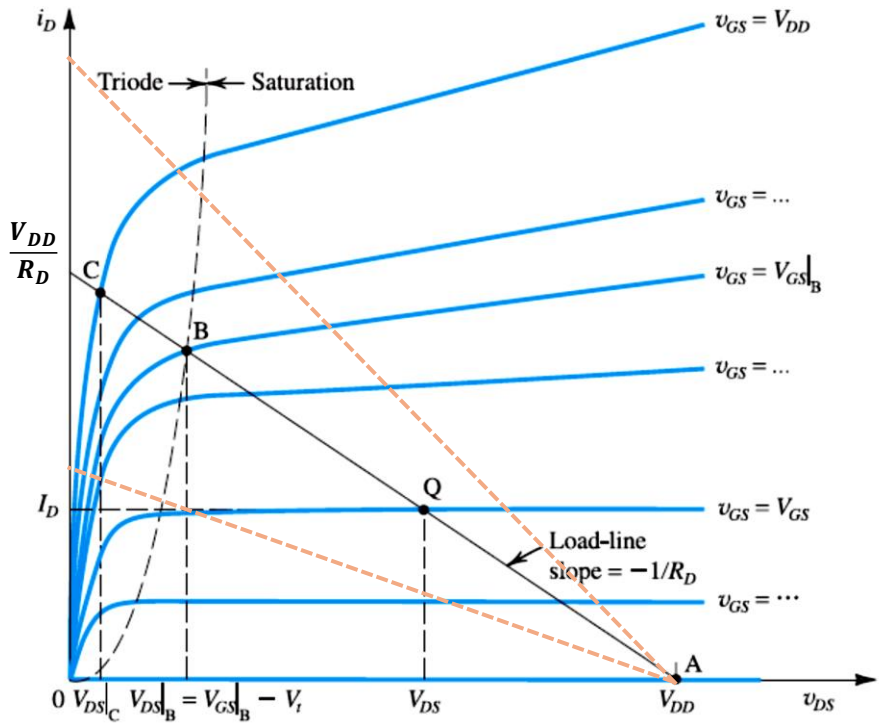


MOSFET BIASING

- Why is biasing required ??
- What Q point signifies
- How to determine the Q point

A,C – SWITCH
A- open switch (off)
C-closed switch (on)





$$V_{DS} = V_{DD} - I_D R_D$$

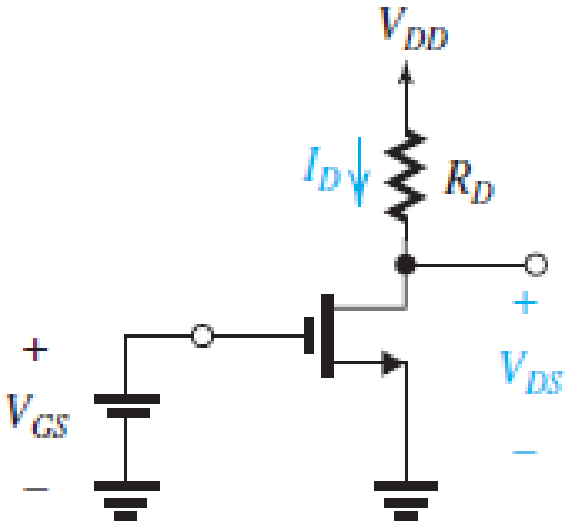
$$I_D = \frac{V_{DD}}{R_D} - \frac{V_{DS}}{R_D}$$

MOSFET BIASING

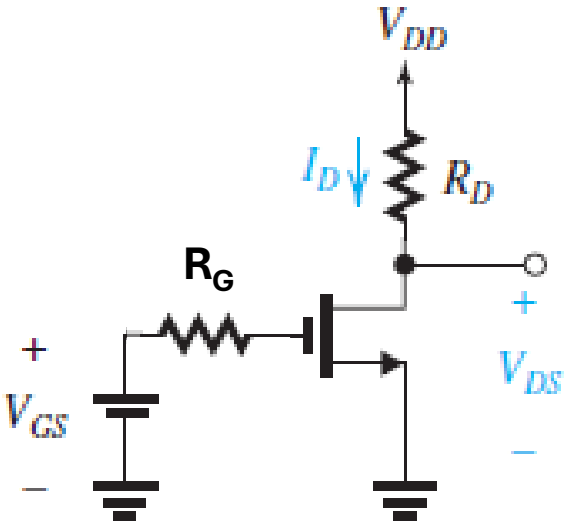
- Biasing means setting up a stable DC operating point (Q-point) for the MOSFET so that it operates in the desired region and amplify small signals linearly

Biasing Techniques

1. Fixed-Bias



$$V_{DS} = VDD - IDRD$$

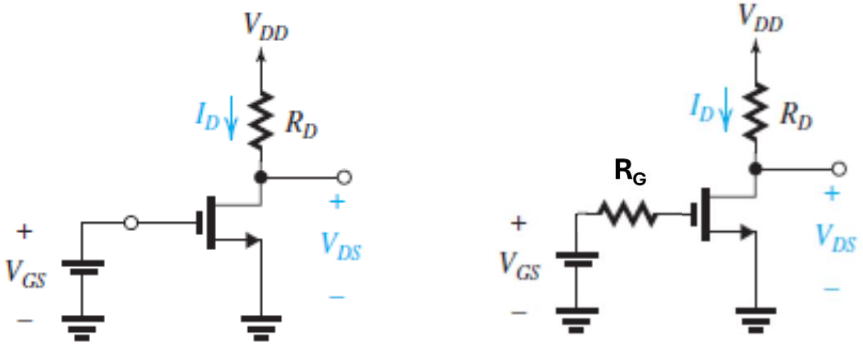


$$I_D = \frac{1}{2}\left[(\mu_n C_{ox})\left(\frac{W}{L}\right)\left((V_{GS} - V_t)^2\right)\right]$$

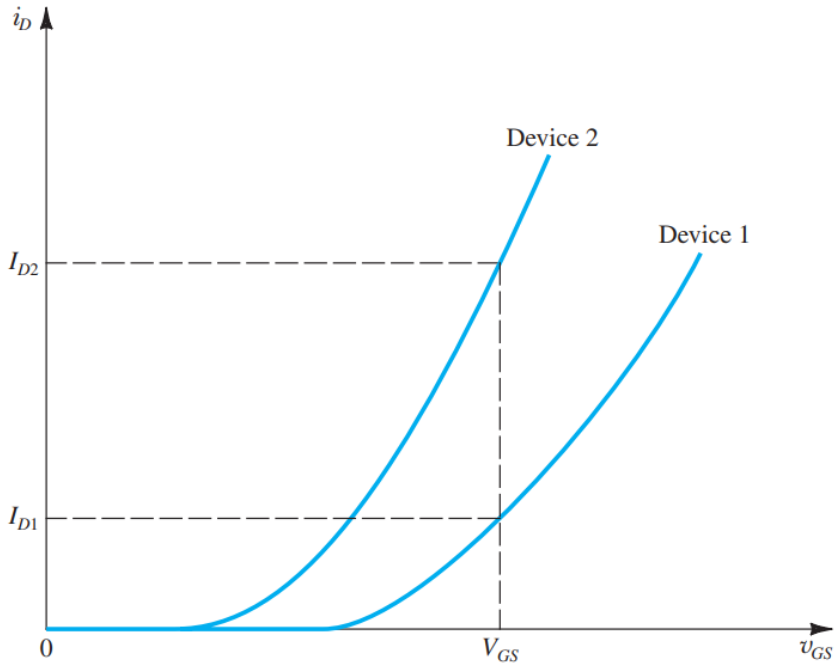
- $V_{GS} = VGS\ varied\ until\ required\ ID\ is\ obtained$
- Once required ID is achieved, VGS is kept constant

MOSFET BIASING

1. Fixed-Bias

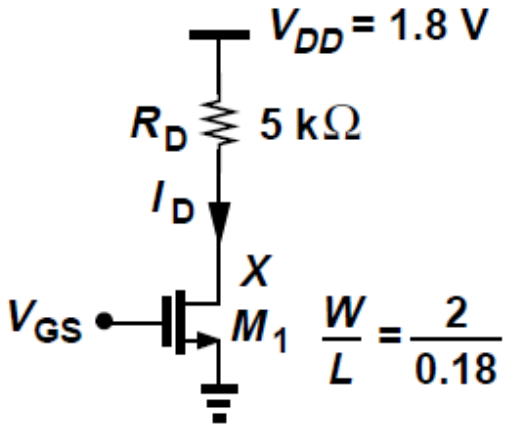


- Simple approach to bias a MOSFET
- not a good approach to bias the MOSFET
- Susceptible to temperature change and device parameters
- μ_n , V_T changes with temperature so I_D changes
- (W/L), VT, Cox changes from device 1 to device 2
- Q point is not stable
- Two MOSFETs will not have same electrical parameters
- This method is not commonly used.



MOSFET BIASING

Calculate the maximum allowable gate voltage in Fig if M1 must remain saturated. $\mu_n C_{ox} = 100\ \mu A/V^2$ and $V_{TH} = 0.4\ V$



MOSFET BIASING

2. Voltage divider-Bias / Resistive divider Bias

How is this better than fixed-bias?

$$I_D = \frac{1}{2}\left[(\mu_n C_{ox})\left(\frac{W}{L}\right)((V_{GS} - V_t)^2)\right]$$

$$I_D = I_G + I_S$$

Here, $I_G = 0$; *hence* $I_D = I_S$

Applying Thevenin's theorem across i/p ckt

$$V_X = \frac{R_2}{R_1 + R_2} V_{DD} = \text{Gate Voltage}$$

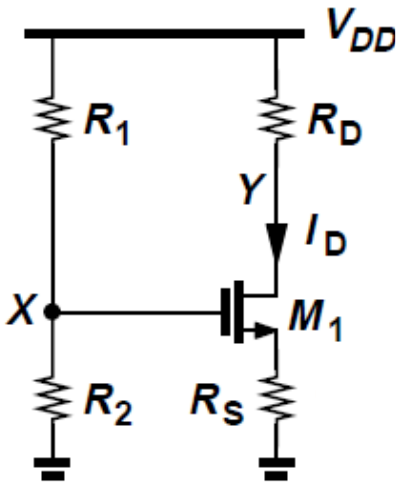
$$R_X = R_1 \parallel R_2$$

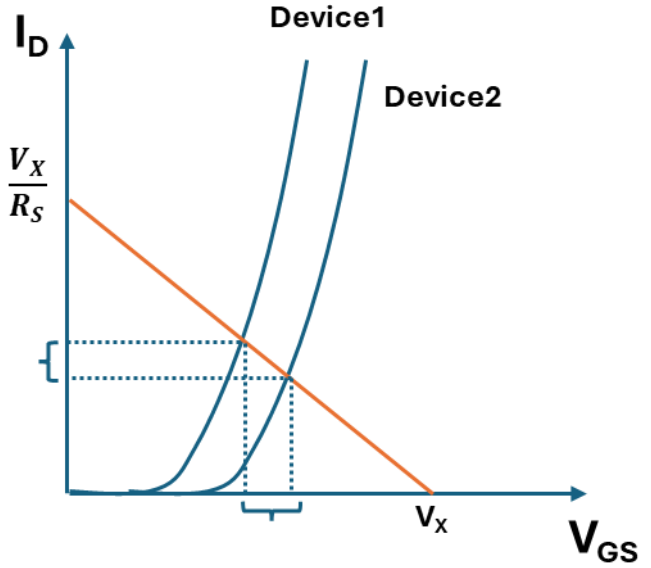
Applying KVL across i/p ckt

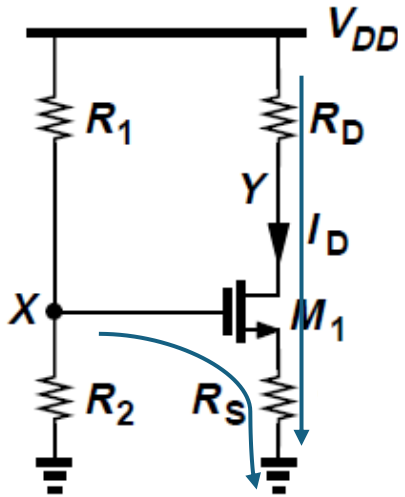
$$V_X - V_{GS} - I_D R_S = 0$$

$$V_{GS} = V_X - I_D R_S$$

$$V_{GS} = V_X - V_S$$







Applying KVL across o/p ckt

$$V_{DD} - I_D R_D - V_{DS} - I_D R_S = 0$$

$$V_{DS} = V_{DD} - I_D (R_D + R_S)$$

$$I_D = \frac{V_{DD} - V_{DS}}{R_D + R_S}$$

MOSFET BIASING

Improved stability:

- By using two resistors to create a voltage divider, the gate voltage is much less sensitive to changes in the MOSFET's parameters.

Stable Q-point:

- The voltage divider creates a stable gate voltage, ensuring the operating point remains relatively constant even with temperature variations or manufacturing differences between MOSFETs.

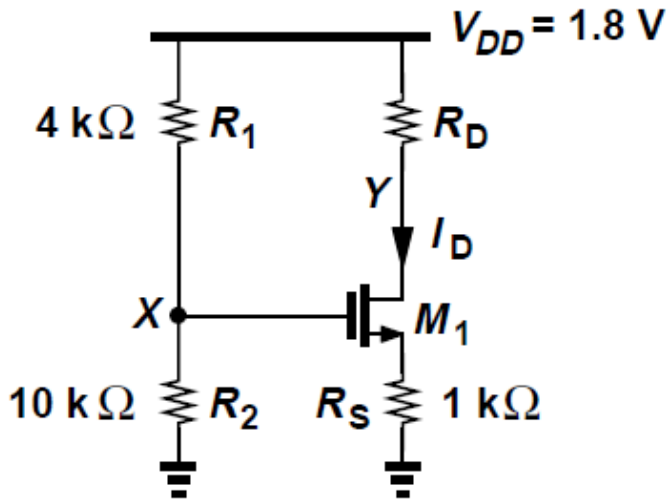
Reduced thermal runaway:

- Fixed bias circuits are prone to thermal runaway, where increased temperature leads to higher drain current, which further increases temperature, and so on. Voltage divider bias helps mitigate this issue.

Suitable for amplifiers:

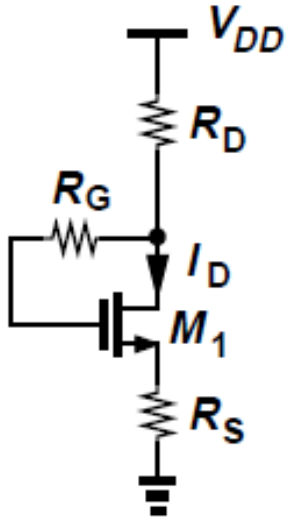
- The stability provided by voltage divider bias makes it a good choice for amplifier circuits where predictable and consistent performance is essential.

2. Voltage divider-Bias / Resistive divider Bias



Determine the bias current of M1 assuming V_{TH} = 0.5 V, $\mu_n C_{ox}$ = 100 μA/V^2 , W/L = 5/0.18, and λ = 0.
What is the maximum allowable value of RD for M1 to remain in saturation?

3. Self-Bias/ Drain-Feedback Bias



$$V_G = V_D$$

$$V_{GS} = VDS = VDD - IDRD$$

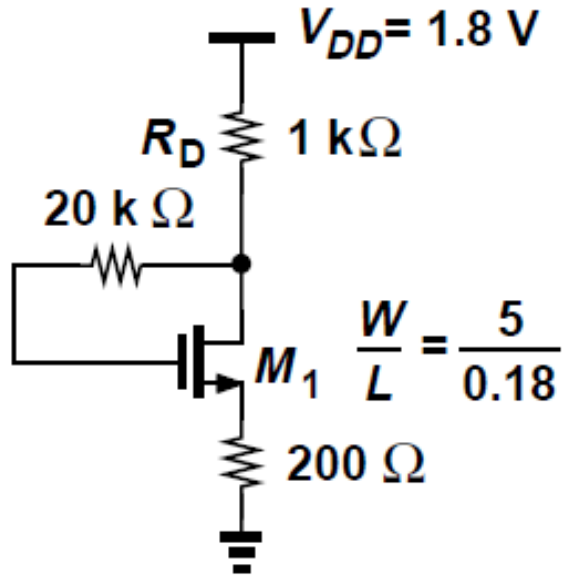
$$I_D = \frac{1}{2}\left[(\mu_n C_{ox})\left(\frac{W}{L}\right)\left((V_{GS} - V_t)^2\right)\right]$$

Feedback in both voltage divider and self bias tries to stabilize the operating point

- Offers higher stability against variations in transistor parameters and temperature compared to voltage divider bias.
- The feedback resistor (RG) creates a loop that automatically adjusts the gate-source voltage (Vgs) to compensate for changes in drain current (Id), helping to maintain a more stable operating point.

MOSFET BIASING

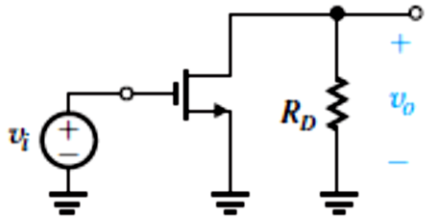
Calculate the drain current ofM1 if $\mu_n C_{ox}$ = 100 A=V2, V_{TH} = 0.5 V, and λ = 0. What value of R_D is necessary to reduce I_D by a factor of two?

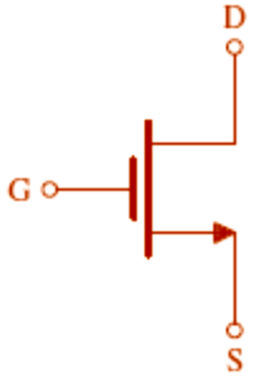


Amplifier Topologies

1. Common-Source Amplifier

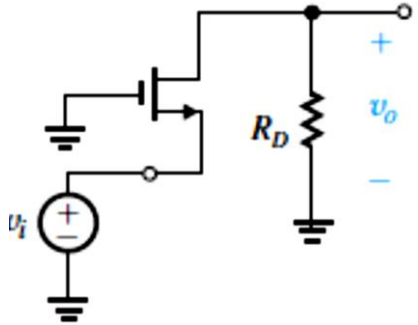
Source common between input and output





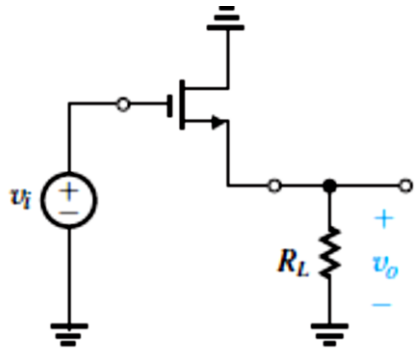
2. Common – Gate Amplifier

Gate common between input and output

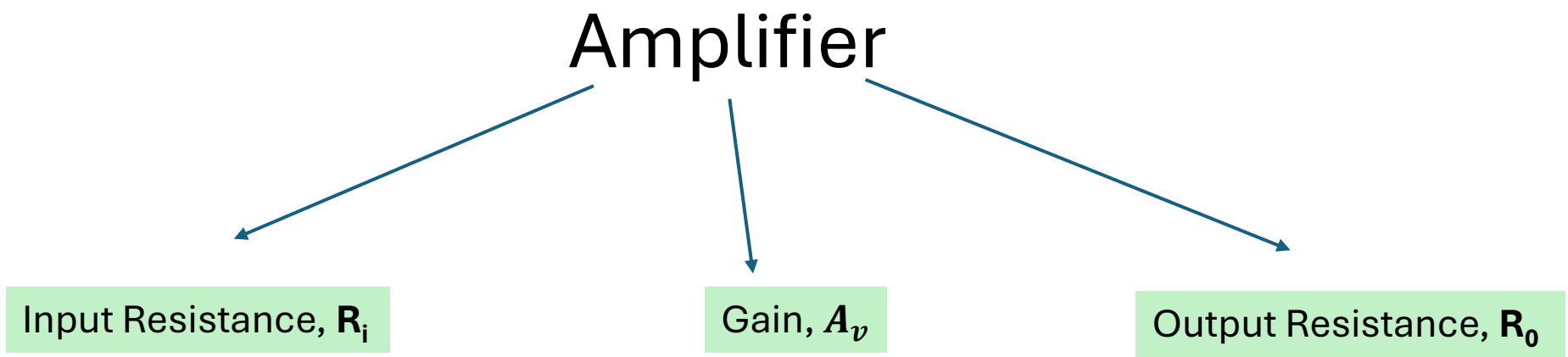


3. Common- Drain Amplifier

Drain common between input and output



Amplifier Topologies



- Resistance seen by the input signal at the **gate terminal** of the MOSFET
- **High $\mathbf{R_i}$** means the amplifier draws almost no current from the signal source

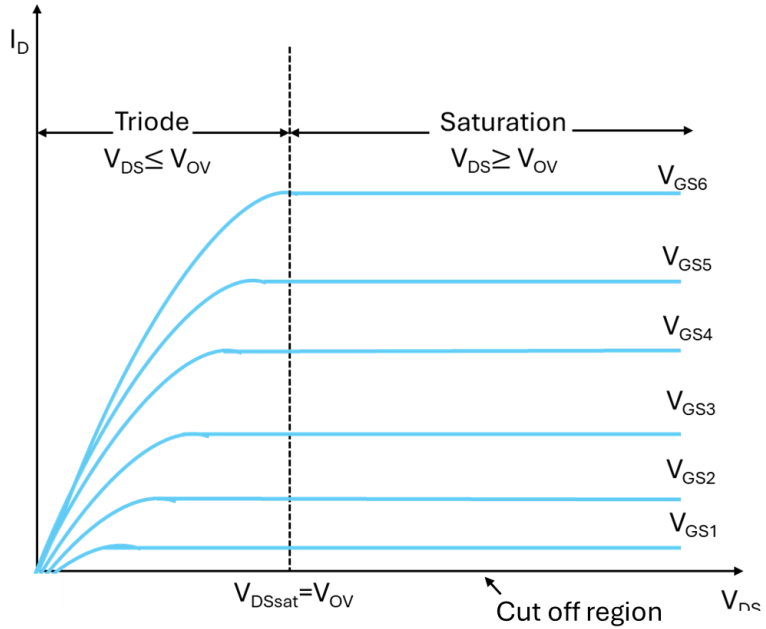
- Ratio of the output voltage to input voltage
- **Shows signal amplification**

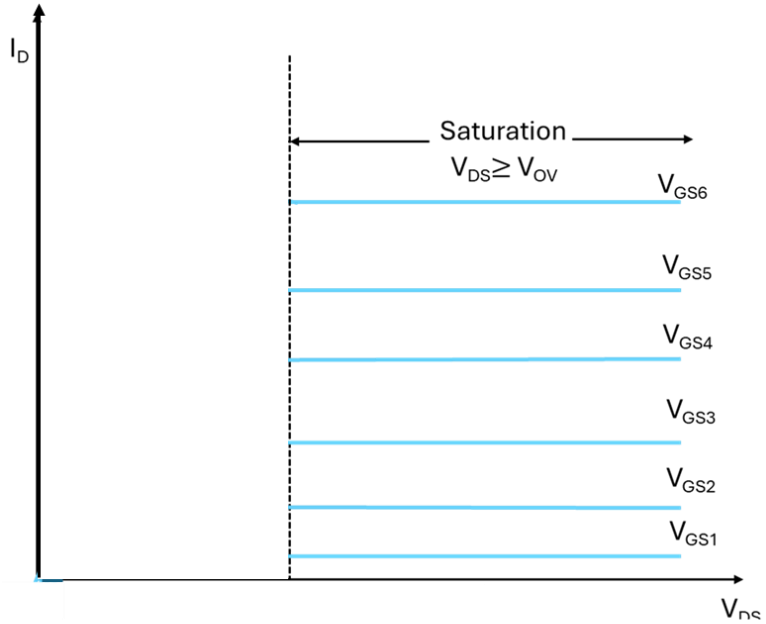
- Resistance seen by the input signal at the **drain terminal** of the MOSFET
- **Low $\mathbf{R_0}$** is better for the power delivery of the load

Amplifier Topologies

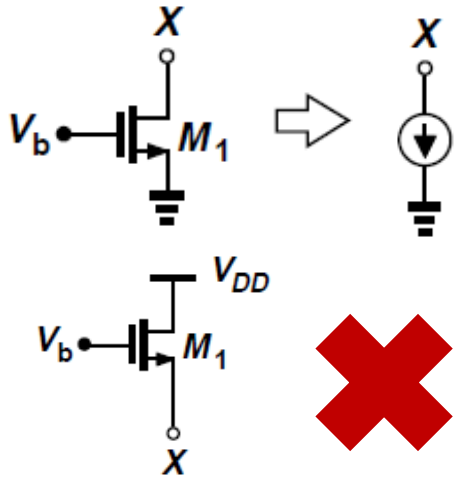
Design Parameter	Voltage Amplifier	Current Amplifier
Input Resistance	**High** – to avoid loading the source and ensure maximum voltage is sensed	**Low** – to allow more current to be drawn from the input source
Output Resistance	**Low** – so that most of the output voltage reaches the load	**High** – to maintain a consistent output current across varying load resistances
Gain Type	**Voltage Gain** – amplifies the voltage level of the input signal	**Current Gain** – increases the current delivered to the load

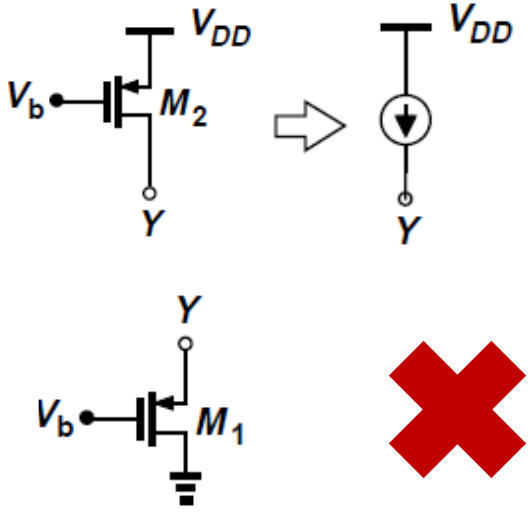
Amplifier Topologies





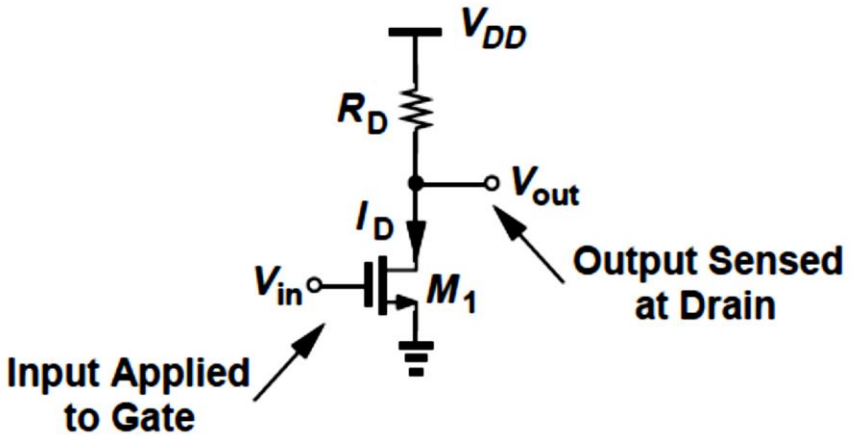
MOSFET in saturation can act as constant current sources

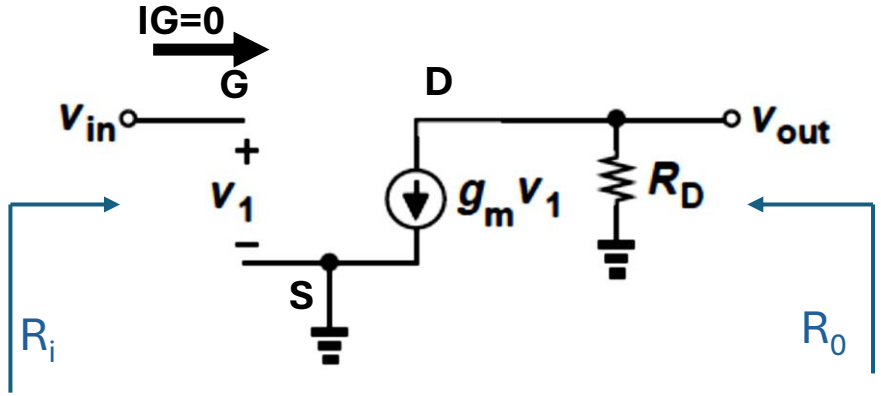




Amplifier Topologies

Common-Source (CS) Amplifier





$$R_i = \infty$$
$$R_0 = R_D$$

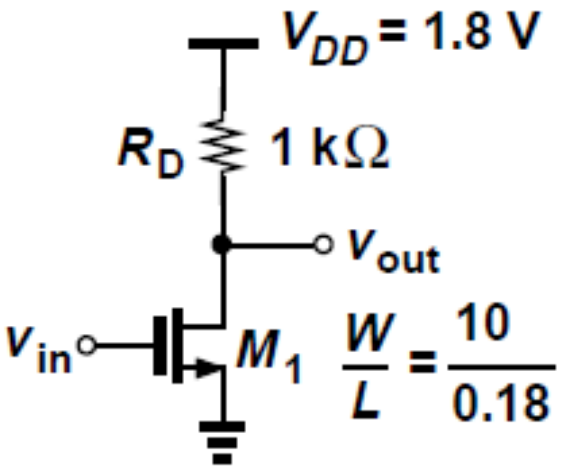
$$V_0 = -g_m V_1 R_D \qquad V_i = V_1$$

Negative sign shows that the **output is 180° out of phase** with the input

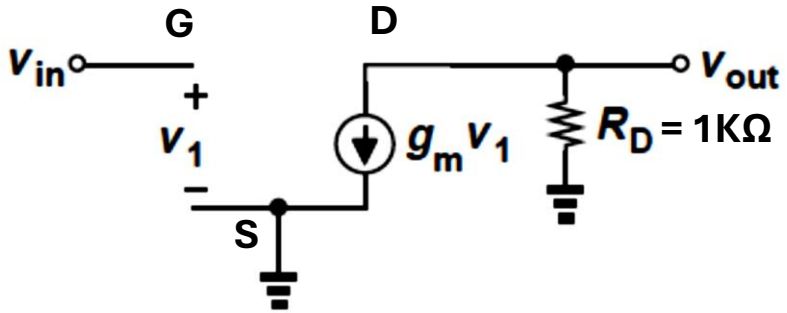
$$A_v = \frac{V_0}{V_i} = -g_m R_D$$

Gain depends on g_m and R_D

Amplifier Topologies



Calculate the small-signal voltage gain of the CS stage if I_D = 1mA, $\mu_n C_{ox}$ =100 μA/V^2, V_{TH} = 0.5 V, and λ = 0.



$$A_v = -g_m R_D$$

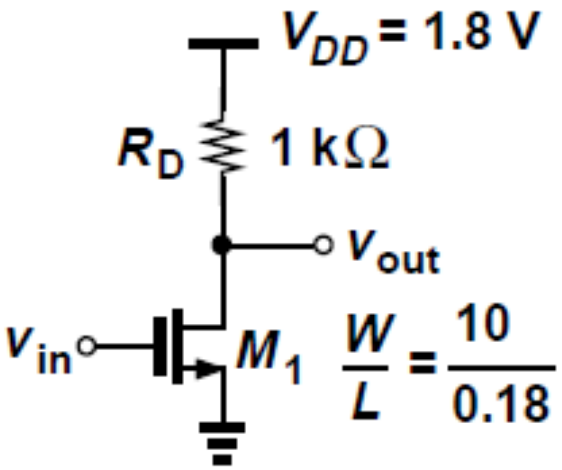
$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}\ const.}$$

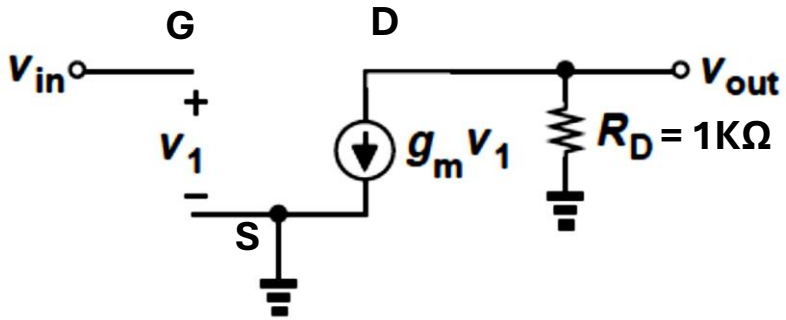
$$g_m = \sqrt{2(\mu_n C_{ox})\left(\frac{W}{L}\right) I_D} = \frac{1}{300}\,\Omega^{-1}$$

$$|A_v| = 3.33$$

What happens when we increase R_D?

Amplifier Topologies





What happens when we increase R_D?

V_{TH} = 0.5 V

$$V_{GS} = V_T + \sqrt{\frac{I_D \, L}{\mu_n C_{OX} W}} = 1.1V$$

$$A_v = -g_m R_D \qquad |A_v| = 3.33$$

$$V_{DS} = V_{DD} - I_D R_D = 1V$$

$$g_m = \sqrt{2(\mu_n C_{ox})\left(\frac{W}{L}\right) I_D} = \frac{1}{300}\,\Omega^{-1}$$

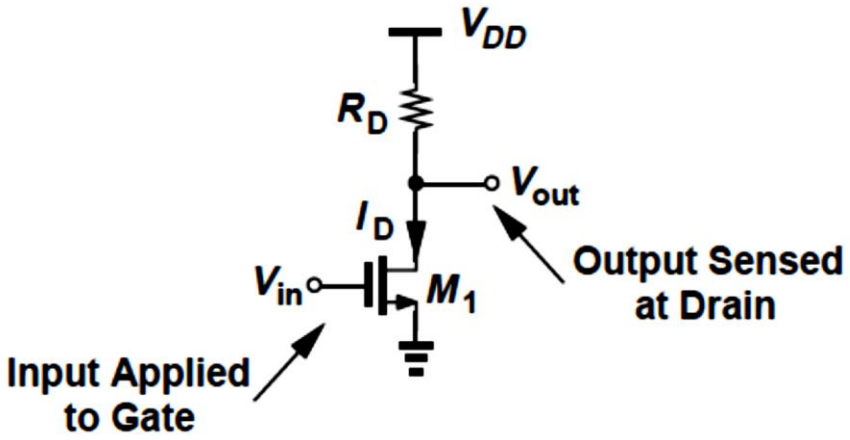
For R_D= 1K, $V_{DS} \geq V_{GS} - VT$

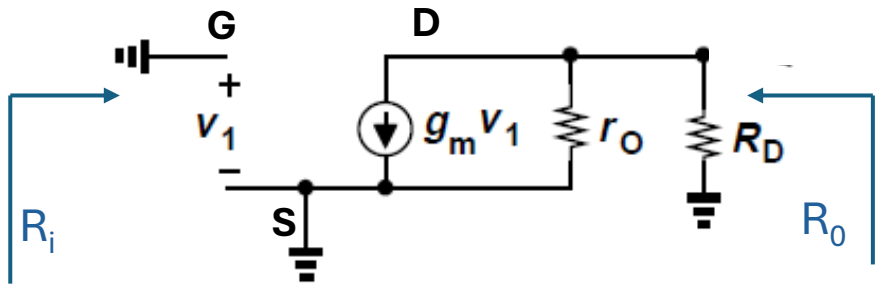
If we increase R_D= 1.2 K, $V_{DS} = 0.6\,V < V_{GS} - VT$

MOSFET may not be in saturation so cannot act as amplifiers

Amplifier Topologies

With Channel Length Modulation





$$R_i = \infty$$
$$R_0 = r_o \| R_D$$

$$V_0 = -g_m V_1 \, (r_o \| R_D) \qquad V_i = V_1$$

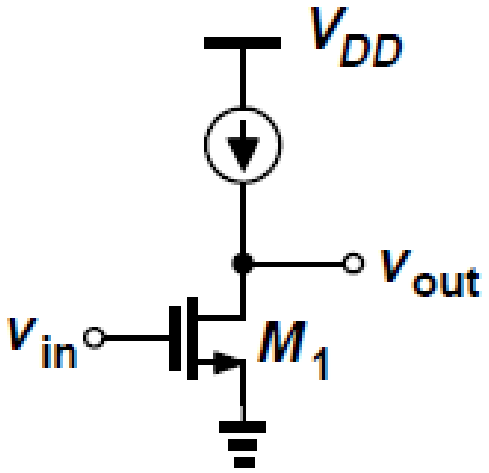
$$A_v = \frac{V_0}{V_i} = -g_m (r_o \| R_D)$$

Resistance as a load :

- *Requires large area, making them unsuitable for compact IC designs.*
- *Difficult to maintain accurate and consistent resistance values.*

Amplifier Topologies

CS Stage With Current-Source Load



$$R_i = \infty$$

$$R_0 = r_o \| R_D \, , R_D = \infty$$

$$R_0 = r_0$$

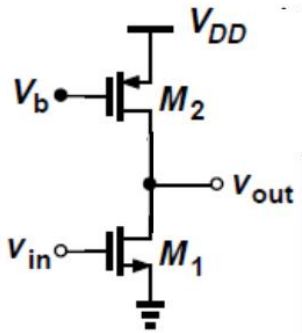
$$V_i = V_1$$

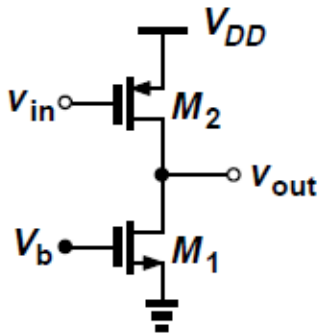
$$V_0 = -g_m V_1 \, (r_0)$$

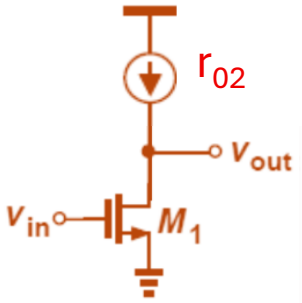
$$\boldsymbol{A_v = \frac{V_0}{V_i} = -g_m \, r_0}$$

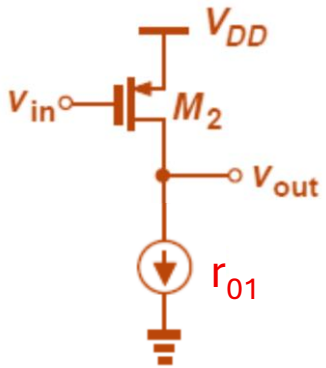
*The trade-off between the voltage gain and the voltage headroom can be relaxed by replacing the **load resistor with a current source***

Amplifier Topologies









$$A_v = -g_{m1}(r_{o1} \| r_{02})$$

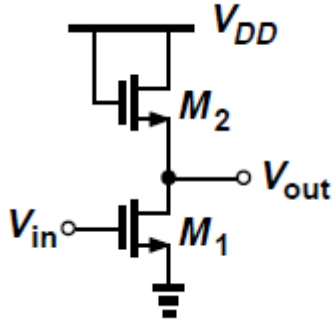
$$A_v = -g_{m2}(r_{o2} \| r_{01})$$

$$R_i = \infty$$

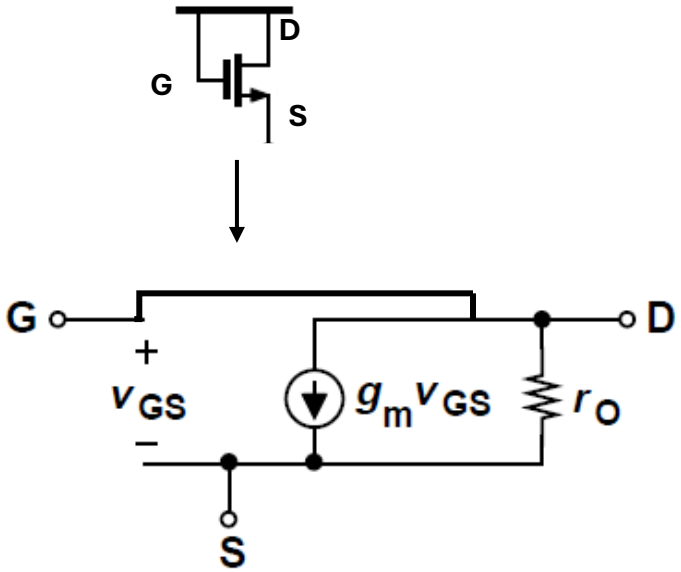
$$R_0 = r_{o1} \| r_{02}$$

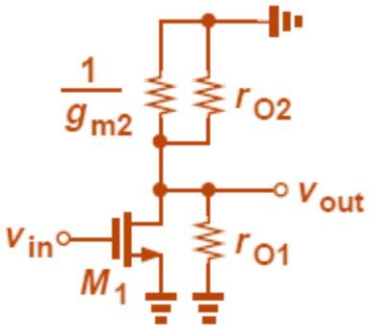
Amplifier Topologies

Diode-connected load



Diode-connection





$$A_v = -g_{m1}\left(\frac{1}{g_{m2}} \| r_{02} \| r_{01}\right)$$

$$V_{GS} = V_{DS}$$

$$i_D = g_m V_{GS} + \frac{V_{DS}}{r_0}$$

$$i_D = (g_m + \frac{1}{r_0}) V_{DS}$$

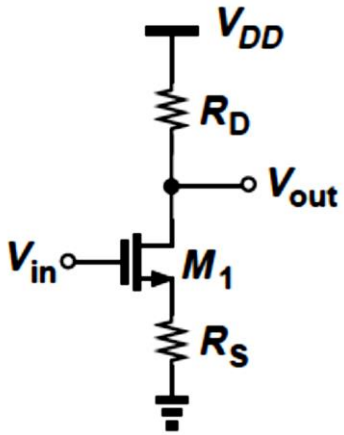
$$\frac{V_{DS}}{i_D} = \frac{1}{\frac{1}{r_0} + g_m} = \frac{\frac{1}{g_m} r_0}{\frac{1}{g_m} + r_0}$$

$$= \left(\frac{1}{g_m} \| r_0\right)$$

If CLM is not considered, then $\boldsymbol{A_v = -\frac{g_{m1}}{g_{m2}}}$

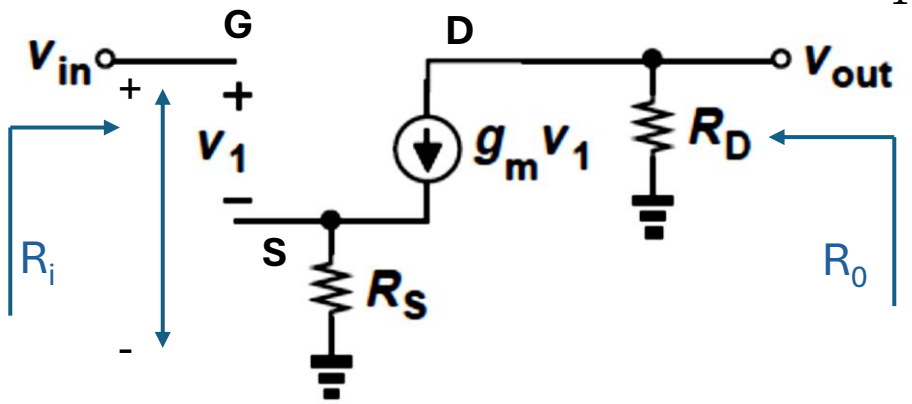
Amplifier Topologies

CS stage with source resistance R_S



$$R_i = \infty$$
$$R_0 = R_D$$

$$V_1 = V_{GS}$$



Negative Feedback

$$V_i = V_1 + g_m V_1 R_S = (1 + g_m R_S) V_1$$

$$V_1 = \frac{V_i}{(1 + g_m R_S)}$$

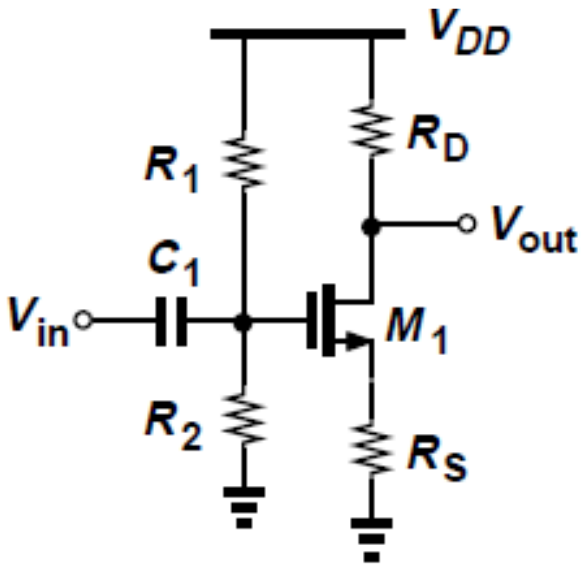
$$V_0 = -g_m V_1 R_D$$

$$A_v = \frac{V_0}{V_i} = \frac{-g_m R_D}{(1 + g_m R_S)}$$

Rs can control the magnitude of the signal V_{GS} and thereby ensure that V_{GS} does not become too large and cause unacceptably high nonlinear distortion.

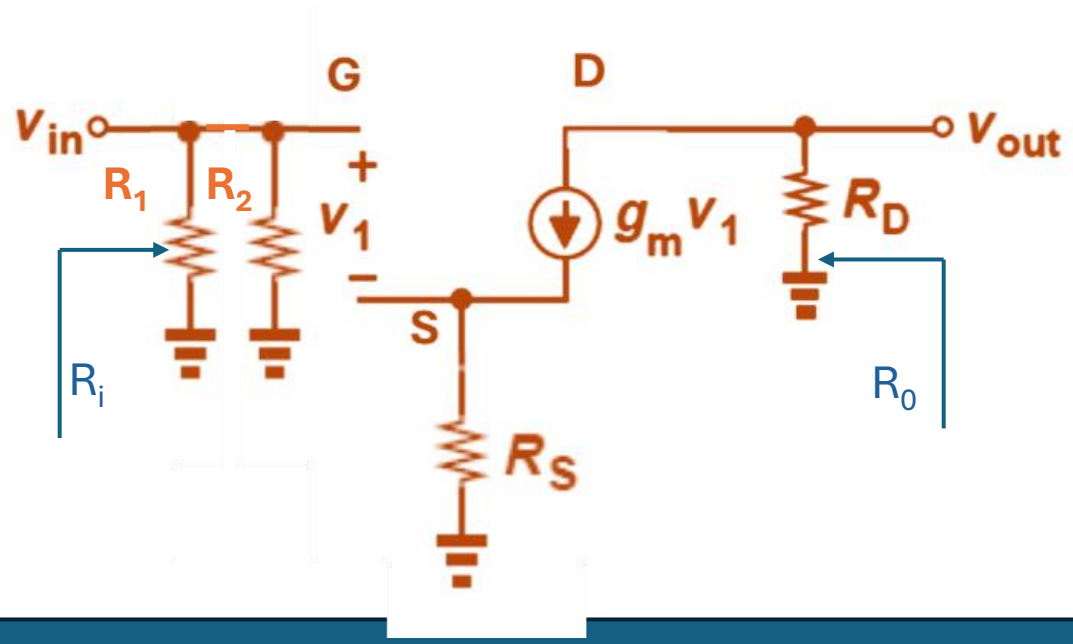
CS With Biasing

Input coupling capacitance



Small signal analysis

1. Short the capacitances.
2. Make V_{DD} to ground.
3. Take the small signal equivalent model

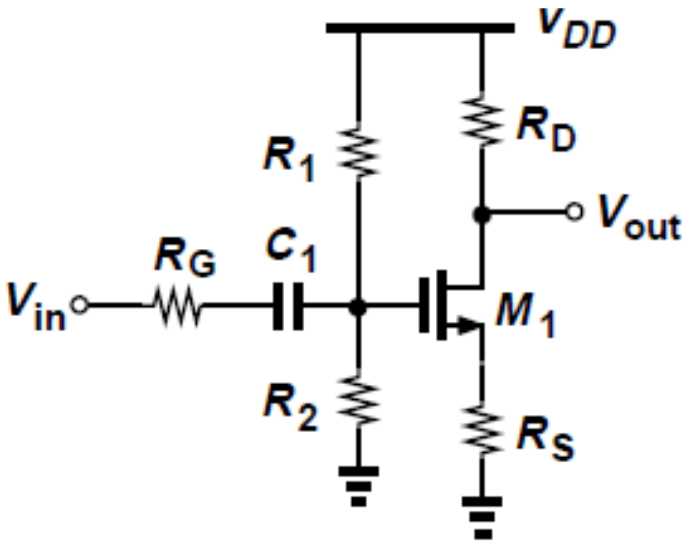


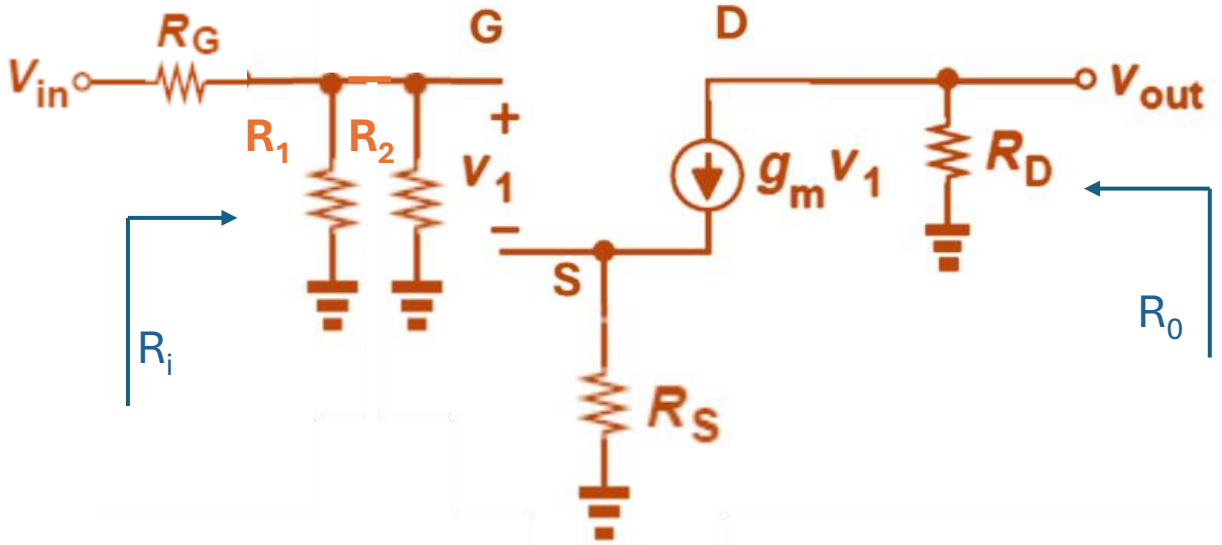
$$R_i = R_1 \| R_2$$
$$R_0 = R_D$$

$$A_v = \frac{-g_m R_D}{(1 + g_m R_S)}$$

Amplifier Topologies

Gate resistance





$$R_i = R_1 \| R_2$$

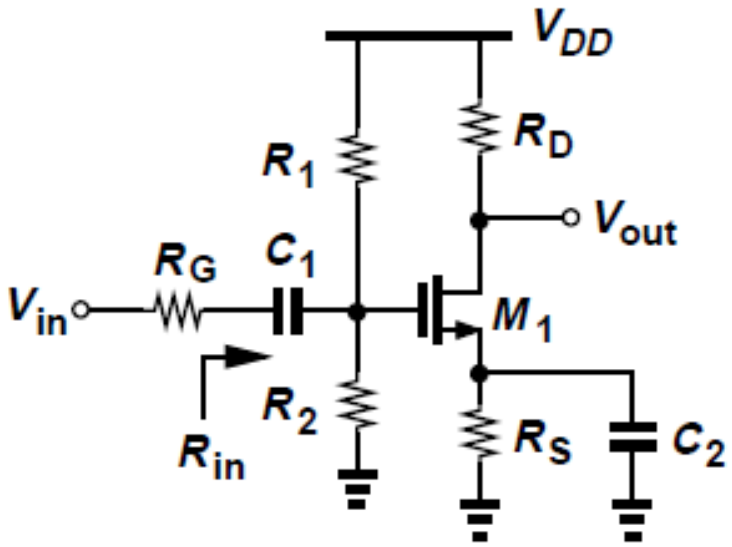
$$R_0 = R_D$$

$$V_i = \frac{R_G + R_1 \| R_2}{R_1 \| R_2}(1 + g_m R_S)$$

$$V_0 = -g_m V_1 R_D$$

$$A_v = \left(\frac{R_1 \| R_2}{R_G + R_1 \| R_2}\right)\frac{-g_m R_D}{(1 + g_m R_S)}$$

Bypass capacitor



Find R_i, R_0 and A_v

Amplifier Topologies

Common-Gate (CG) Amplifier

Input Impedance

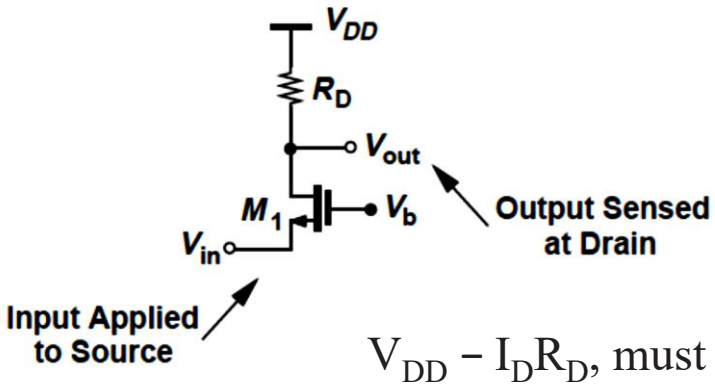
$$R_{in} = \frac{V_{in}}{I_{in}}$$

$$I_{in} + g_m V_1 = 0$$

$$I_{in} = -g_m V_1$$

$$V_{in} = -V_1$$

$$R_{in} = \frac{1}{g_m}$$

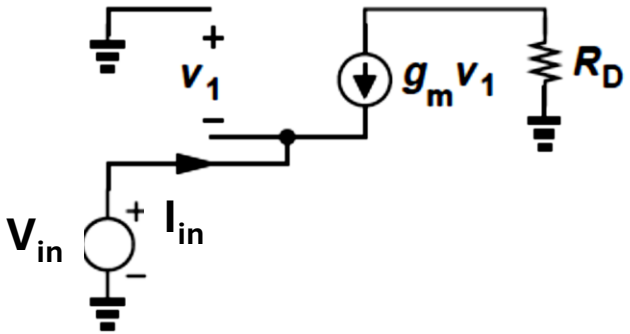


Output Impedance

$$R_o = R_D$$

$$V_0 = -g_m V_1 R_D$$

$V_{DD} - I_D R_D$, must remain above $V_b - V_{TH}$ to ensure M1 is saturated.



Current gain, $A_v = \frac{I_o}{I_i} = 1$

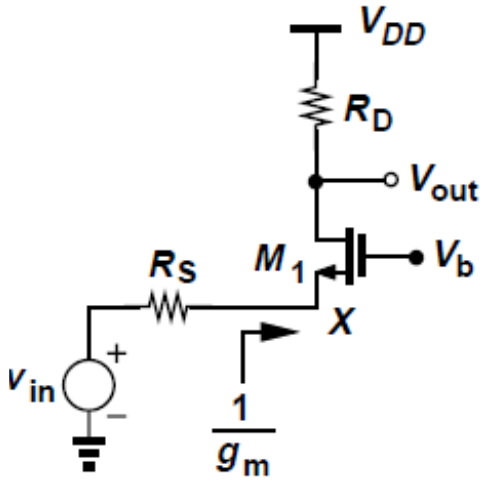
Gain

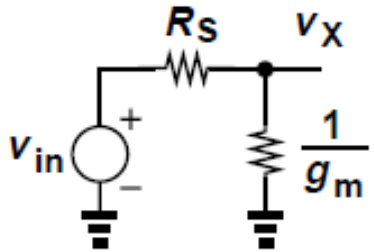
$$A_v = g_m R_D$$

Gain depends on R_D, suffers from voltage headroom-gain trade-offs like that of the CS amplifier

Amplifier Topologies

CG Amplifier with source resistance





$$V_X = -V_1 \qquad V_0 = -g_m V_1 R_D$$

$$R_{in} = \frac{1}{g_m} \qquad R_o = R_D$$

$$V_X = \frac{\frac{1}{g_m}}{R_S + \frac{1}{g_m}} V_{in}$$

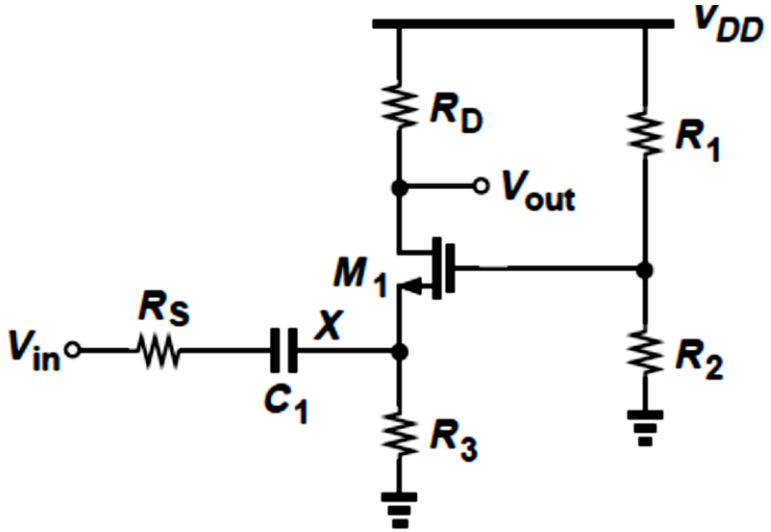
$$V_X = \frac{1}{1 + g_m R_S} V_{in}$$

$$A_v = \frac{V_0}{V_{in}} = \frac{g_m R_D}{1 + g_m R_S} = \frac{R_D}{R_S + \frac{1}{g_m}}$$

Gain same as CS amplifier with source resistance

Amplifier Topologies

The voltage divider consisting of R1 and R2 neglected because Gate is grounded



$$V_X = -V_1 \qquad V_0 = -g_m V_1 R_D$$

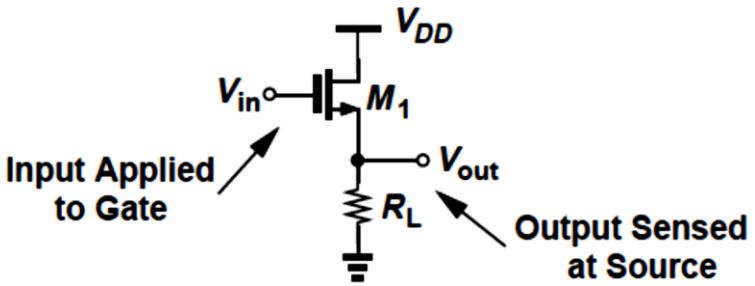
$$R_{in} = \frac{1}{g_m} \qquad R_o = R_D$$

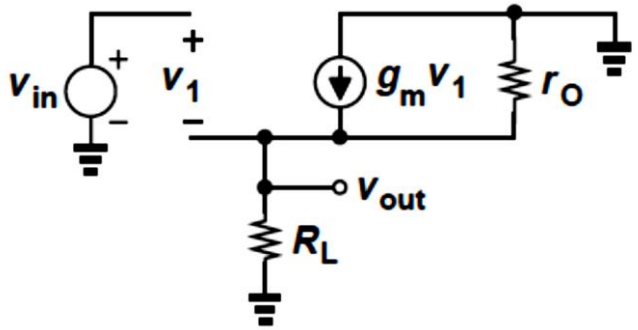
$$V_X = \frac{\frac{1}{g_m} \| R_3}{R_S + \frac{1}{g_m} \| R_3} V_{in}$$

$$A_v = \frac{\frac{1}{g_m} \| R_3}{R_S + \frac{1}{g_m} \| R_3} (g_m R_D)$$

Amplifier Topologies

Common-Drain (CD) Amplifier





Output is sensed at source, hence, also known as **Source Follower**

$$V_{in} = V_1 + V_{out}$$

$$V_{out} = g_m V_1 (r_0 \| R_L)$$

$$V_{in} = V_1 (1 + g_m (r_0 \| R_L))$$

$$A_v = \frac{g_m (r_0 \| R_L)}{1 + g_m (r_0 \| R_L))}$$

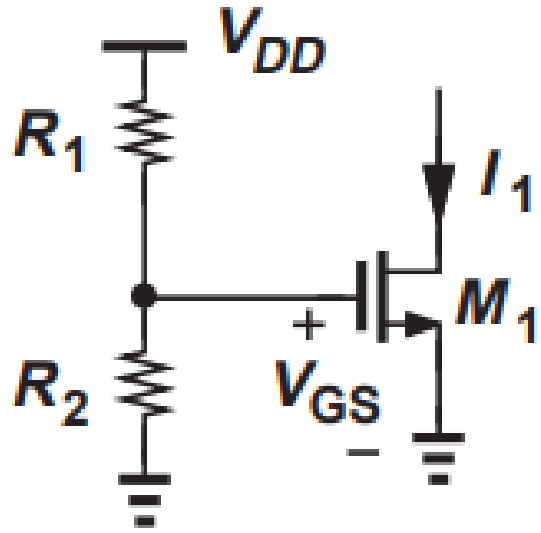
Current mirror

- The biasing techniques studied for MOS amplifiers prove inadequate for high-performance microelectronic circuits.
- For example, the bias current of CS stages is a function of the supply voltage—a serious issue because in practice, this voltage experiences some variation.
- The rechargeable battery in a cellphone or laptop computer, for example, gradually loses voltage as it is discharged, thereby mandating that the circuits operate properly across a range of supply voltages.
- Another critical issue in biasing relates to ambient temperature variations.
- A cellphone must maintain its performance at −20°C in Finland and +50 °C in Saudi Arabia.

A MOS current source biased by means of a resistive divider suffers from dependence on VDD and temperature.

$$I_1 = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$$

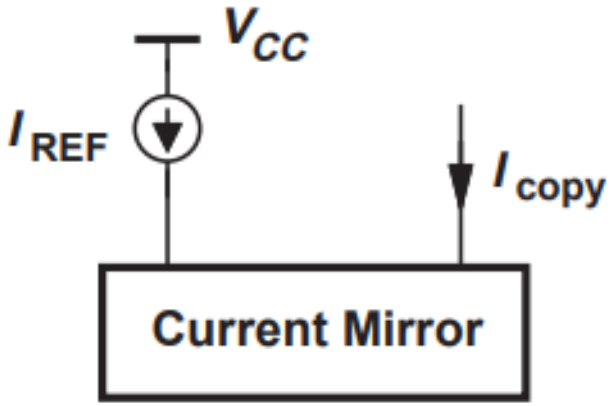
$$= \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left(\frac{R_2}{R_1 + R_2}V_{DD} - V_{TH}\right)^2.$$



Since both the mobility and the threshold voltage vary with temperature, I1 is not constant even if VGS is.

Current mirror

- An elegant method of creating supply- and temperature-independent voltages and currents exists and appears in almost all microelectronic systems.
- Called the “bandgap reference circuit” and employing several tens of devices.



- In order to avoid supply and temperature dependence, a bandgap reference can provide a “golden current” while requiring a few tens of devices.
- We must therefore seek a method of “copying” the golden current without duplicating the entire bandgap circuitry. Current mirrors serve this purpose.
- Icopy = IREF

Current mirror

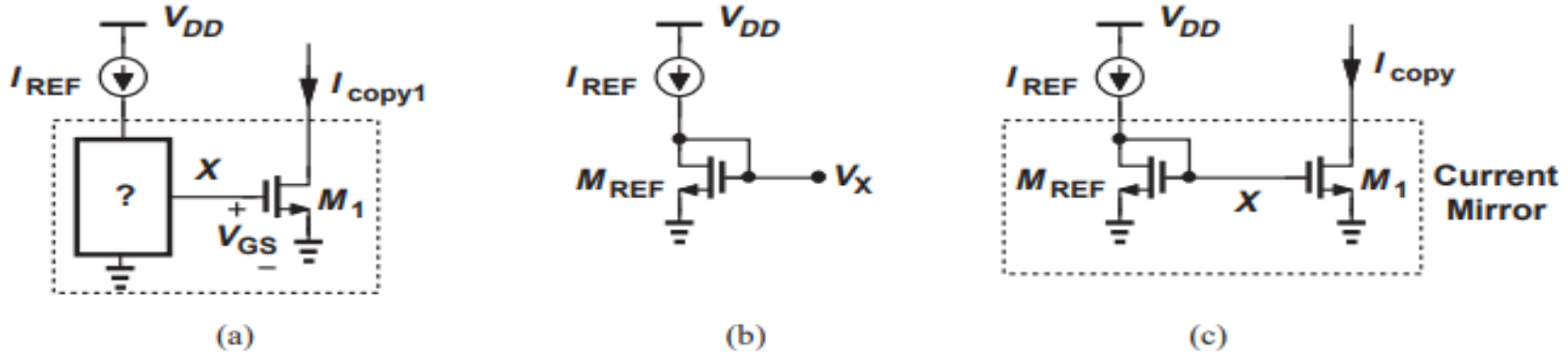


Figure 9.35 (a) Conceptual illustration of copying a current by an NMOS device, (b) generation of a voltage proportional to square root of current, (c) MOS current mirror.

The black box must generate VX such that

$$\frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_X - V_{TH1})^2 = I_{REF}$$

where channel-length modulation is neglected. Thus, the black box must satisfy the following input (current)/output (voltage) characteristic:

$$V_X = \sqrt{\frac{2I_{REF}}{\mu_n C_{ox}\left(\frac{W}{L}\right)_1}} + V_{TH1}.$$

Current mirror

- That is, it must operate as a “square-root” circuit.
- A diode-connected MOSFET provides such a characteristic [Fig. 9.35(b)], thus arriving at the NMOS current mirror depicted in Fig. 9.35(c).
- The drain currents of the two transistors can be expressed as:

$$I_{D,REF} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_{REF}(V_X - V_{TH})^2$$

$$I_{copy} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_{1}(V_X - V_{TH})^2,$$

where the threshold voltages are assumed equal. It follows that

$$I_{copy} = \frac{\left(\frac{W}{L}\right)_1}{\left(\frac{W}{L}\right)_{REF}} I_{REF},$$

which reduces to $I_{copy} = I_{REF}$ if the two transistors are identical.